Exam 1 –
Math 401—Calculus II
Spring 2021

Name_____
Date_____
Instructor: Rocio Owens

• *You may use a scientific calculator or a TI-83/84/86 calculator on this exam.* *TI89 or TI-nspire models are not allowed.*
• *For full credit, fractional answers must be reduced and decimal answers must be rounded correctly.*
• *Each problem is worth 10 points, unless otherwise specified, for a total of 100 points.*
• *Please put each answer in the blank when provided so I'm not looking for it. Good luck!*

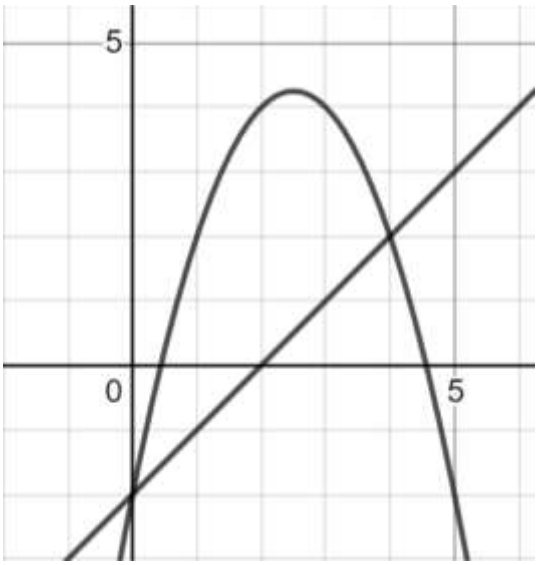
1) (10 pts each) Use the **general power rule or u-substitution** to integrate the following:

a) $\int \frac{6 - 3x + 8x^5}{2x^2}\,dx$ **1a) _____**

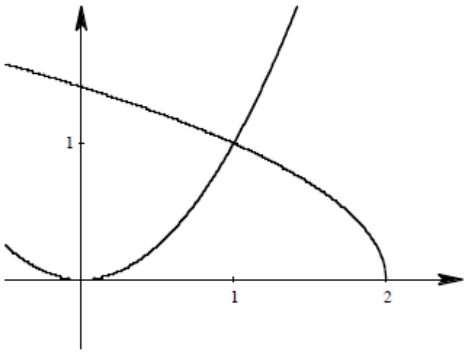
b) $\int \frac{x^2 \ln(x^3+2)}{x^3+2}\,dx$ 1b) _____

c) $\int \frac{x^5}{\sqrt[4]{1+x^3}}\,dx$ 1c) _____

2) Find the area of the region that is bounded by the graphs of $y = x - 2$ and $y = 5x - x^2 - 2$.

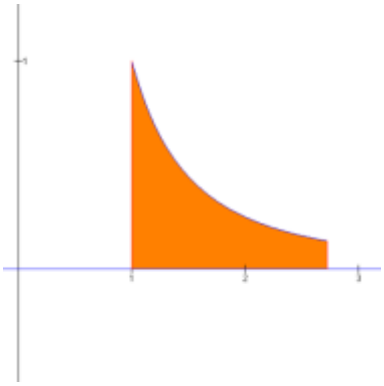


3) The region in the first quadrant bounded by the x-axis, the graph of $y = x^2$ and the graph of $y = \sqrt{2 - x}$ is rotated about the x-axis. SET-UP an integral that would compute the resulting volume.

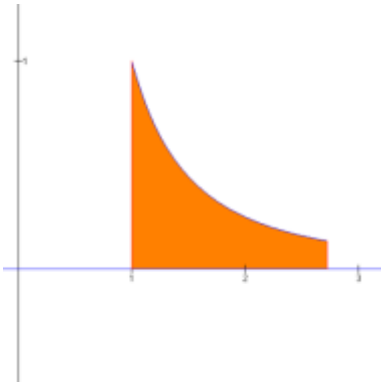


4) (10 each) Let R be the region bounded by the graphs of $y = \frac{1}{x^2}$, , x = 1, x = e and y = 0.

a) Use the shell method to find the volume of the solid generated when R is revolved about the y-axis. SET-UP only.



b) Use the disk/washer method to **Set-Up** an integral to find the volume of the solid generated when R is revolved about the line y = -1: SET-UP only.



5) A cable that weighs 2lbs/ft is used to lift 800 lbs of coal up a mine shaft 500 ft deep. Find the work done in lifting the coal to the top. Recall: W = Force*Displacement

6) Suppose that 2 Joules of work is needed to stretch a spring from its natural length of 30 cm to a length of 42cm. How much work is needed to stretch the spring from 35 cm to 40 cm?
Recall by Hooke's Law F = kx

7) Use the general slicing method to find the volume of a solid with a semicircular base of radius 10 whose cross sections perpendicular to the base are squares. SET-UP only.

8) (Extra Credit) A cylindrical propane tank (r = 4 m, h =10 m) lies on its side and is filled with propane (density = 493kg/m^3) to a depth of 2m. SET UP an integral for the total work required to pump all of the propane to an outlet valve at the top of the tank.
$\int_a^b \rho \cdot g \cdot (Area\ of\ crossection)(distance\ out\ of\ tank)dy\ (or\ dx)$
Where ρ = density = 493 kg/m^3 and g = 9.8 m/s^2.

